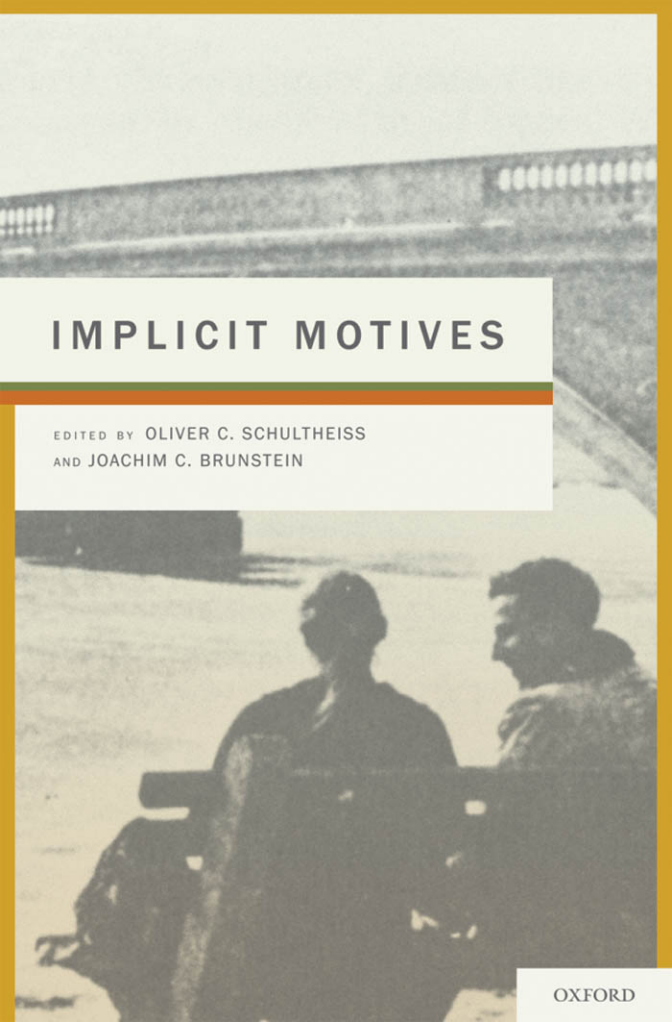




IMPLICIT MOTIVES

EDITED BY OLIVER C. SCHULTHEISS
AND JOACHIM C. BRUNSTEIN

OXFORD

Implicit Motives

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Contents

Contributors vii

Introduction

Oliver C. Schultheiss and Joachim C. Brunstein ix

SECTION 1 MOTIVE SYSTEMS 1

1 Power Motivation

Eugene M. Fodor 3

2 The Achievement Motive: A Review of Theory and Assessment
of *n* Achievement, Hope of Success, and Fear of Failure

Joyce S. Pang 30

3 The Duality of Affiliative Motivation

Joel Weinberger, Tanya Cotler, and Daniel Fishman 71

4 Activity Inhibition

Thomas A. Langens 89

SECTION 2 ASSESSMENT OF IMPLICIT MOTIVES 117

5 Content Coding Methods in Implicit Motive Assessment: Standards of
Measurement and Best Practices for the Picture Story Exercise

Joyce S. Pang 119

6 Assessing Individual Differences in Achievement Motivation with the
Implicit Association Test: Predictive Validity of a Chronometric Measure
of the Self-Concept “Me = Successful”

Joachim C. Brunstein and Clemens H. Schmitt 151

7 Computer-Based Modeling, Assessment, and Coding of Implicit Motives

Virginia Blankenship 186

SECTION 3 BASIC CONCEPTS AND PROCESSES 209

- 8 Learning and Memory Correlates of Implicit Motives
Michael Bender and Barbara A. Woike 211
- 9 Properties of Motive-Specific Incentives
Steven J. Stanton, Julie L. Hall, and Oliver C. Schultheiss 245
- 10 Biopsychological and Neural Processes of Implicit Motivation
Julie L. Hall, Steven J. Stanton, and Oliver C. Schultheiss 279
- 11 Factors That Influence the Relation Between Implicit and Explicit Motives:
A General Implicit–Explicit Congruence Framework
*Todd M. Thrash, Scott E. Cassidy, Laura A. Maruskin, and
Andrew J. Elliot* 308
- 12 Implicit Motives and Explicit Goals: The Role of Motivational Congruence
in Emotional Well-Being
Joachim C. Brunstein 347
- 13 Implicit Motives: A Look from Personality Systems Interaction Theory
Nicola Baumann, Miguel Kazén, and Julius Kuhl 375

SECTION 4 INTERDISCIPLINARY AND APPLIED ASPECTS 405

- 14 Political and Historical Consequences of Implicit Motives
David G. Winter 407
- 15 Research on Implicit Motives Across Cultures
Jan Hofer 433
- 16 Clinical Implications of Implicit Motives
Joel Weinberger, Tanya Cotler, and Daniel Fishman 468
- 17 Competencies as a Behavioral Manifestation of Implicit Motives
Richard E. Boyatzis and Stephen P. Kelner, Jr. 488
- 18 Motive Training and Motivational Competence
Falko Rheinberg and Stefan Engeser 510

Index 549

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Introduction

This volume continues a loose series of edited books (Atkinson, 1958; Smith, 1992) dedicated to the examination of human motivation and behavior through the lens of implicit motives, enduring non-conscious needs that drive humans' behavior toward the attainment of specific classes of incentives. In the following, we will sketch out a brief history of research on implicit motives, portray the basic principles guiding implicit motive research, and provide an overview of the chapters featured in this book.

The Excitement of the Early Years : 1948 to 1961

Research on implicit motives started when in the late 1940s David McClelland, who was then a new faculty member at Wesleyan University, teamed up with graduate student John Atkinson to measure motivational needs in humans. They decided to start with hunger motivation as a model need system but did not want to rely on people's introspective reports of hunger because they doubted the validity of such self-reports. McClelland (1984) later traced his doubts about self-report to his observations as a young man who noted the striking difference between the values people avowed to in church on Sundays and their actual behavior throughout the rest of the week. "I never put much faith in what people say their values are on questionnaires, because I don't believe that these statements bear very much relationship to what they in fact do or even to the values that *implicitly* guide their lives. This also gave me a strong belief in the reality of unconscious values or motives, which were obviously affecting what they did in ways that were quite unknown to themselves" (p. 4, italics in original).

McClelland and Atkinson were looking for a way to assess the need for food that bypassed research participants' introspective reports on how hungry they felt. According to Atkinson (in Winter, 1998), the decision to measure hunger motivation with a variant of the Thematic Apperception Test (TAT), Morgan & Murray's (1935) device to tap into unconscious needs, was due to a lucky coincidence: "We were talking about this first experiment in Dave's office. Bob Knapp walked from one side to the other and he said, 'Why don't you try the TAT?', and then walked out the door." (Winter, 1998, p.139)

The decision to use the TAT was due to serendipity; the rest is history. McClelland and Atkinson's original work on the effects of food deprivation on TAT story content was extremely promising (Atkinson & McClelland, 1948). Individuals who had fasted for 16 hours wrote stories about pictures suggestive of food that dealt with the procurement of food and with deprivation states; individuals who had just had breakfast did not imbue their stories with this kind of imagery. Thus, Atkinson and McClelland had used an experimental manipulation of a motivational need (food deprivation versus no deprivation) and studied its effect on the themes research participants wrote about in their stories. In later research, the themes identified in experimental motive arousal studies of this kind were codified and used to assess individual differences in motivational needs in individuals who had been tested under neutral conditions. This was based on the assumption that individuals who showed a lot of motive imagery in their stories in the absence of situational motive arousal must be chronically high in this motivational need whereas individuals with little motive imagery are not.

Despite the initial success with identifying imagery differences in stories written by hungry and full research participants, hunger as a motivational need was soon abandoned in favor of the need to achieve because it was felt that this need was more relevant for individuals' success in specific jobs and life in general. Within only a handful of years, McClelland, Atkinson, and their collaborators conducted ground-breaking research on the assessment of the need for achievement (or *n* Achievement, as it was named to adhere to Murray's, 1938, original terminology). The harvest of this research was published as *The Achievement Motive* (McClelland, Atkinson, Clark, & Lowell, 1953), a book that remains highly readable and thought-stimulating to this day and contains many ideas and findings whose implications still await thorough study.

The years after the publication of *The Achievement Motive* saw a rapid proliferation of the measurement approach to motivation introduced by McClelland and Atkinson. Enthusiasm for the new motivational concepts and measurement approach quickly spread beyond the confines of Harvard University, where David McClelland received an appointment in the early 1950s, and the University of Michigan, where John Atkinson continued his work after graduating from Wesleyan. In quick succession, TAT-based

measures of other motivational needs were developed, such as the needs for power, affiliation, and sex, and validated in studies on task performance, perception, attitudes, developmental processes, and other validity criteria. Researchers also dedicated considerable effort to understanding the picture story coding methods used for the assessment of motives. Many of these studies were published or reprinted in the first edited volume dedicated solely to theory and research dealing with implicit motives, Atkinson's (1958) *Motives in Fantasy, Action, and Society*.

The early years also portended what would come to be a hallmark of implicit motive research, namely the application of motivational measures and concepts beyond the confines of psychological science. Soon after the initial work on the assessment and validation of the need for achievement was completed, McClelland started to explore the usefulness of this motive for the explanation of economic phenomena and their cultural precursors. These ideas were formulated for the first time in a contribution to the Nebraska Symposium on Motivation series (McClelland, 1955) and culminated in the publication of *The Achieving Society* (McClelland, 1961), an exciting intellectual tour-de-force through history, mythology, religion, sociology, and particularly economics. In the book, McClelland linked Protestant values to child-rearing practices that fostered high achievement motivation and, as a consequence, entrepreneurial activity. He also demonstrated for several different countries and historical eras that periods of increases in collective achievement motivation preceded periods of economic growth. *The Achieving Society* represents the crowning accomplishment of the field's early years, pushing the boundaries of what motivational science (and psychology more generally) can be applied to and account for, and showcasing the impressive validity and predictive power of content-coding methods for the assessment of motivational needs.

Consolidation, Confusion, and Conflict: 1961 to 1989

For roughly the next three decades, work on implicit motives branched out further, first continuing to gain interest among researchers, but then slipping into stagnation in the later years. The factors that contributed to its initially increasing popularity included Atkinson's development of a rigorous and highly influential theory of achievement motivation that was originally published in 1957 but had its main impact on psychology and other social sciences in the 1960s and 1970s. Atkinson later took theorizing a big step further by developing, with David Birch, the dynamics of action theory (Atkinson & Birch, 1970), a highly advanced and complex account of the laws governing the ebb and flow of motivation and the change from one activity to the next.

Meanwhile, McClelland and his associates intensified their efforts to apply motivational concepts to real-world phenomena in business settings and economic development. One such effort was the attempt to train small business owners in an entire Indian town to think and act like a person high in the need for achievement and thereby to improve the local economy (McClelland & Winter, 1969). Another success story of the second phase in the field's history was the emerging realization that implicit motives were associated with specific psychophysiological responses and health outcomes. Steele (1973), in his pioneering dissertation, was able to document a link between aroused power motivation and excretion of sympathetic catecholamines, as measured in urine levels. With a time lag of 10 years, this discovery led to an intensive and productive exploration of the role of implicit motives in health and disease and the physiological mechanisms involved in it (for a summary, see McClelland, 1989).

Despite these successes, the enthusiasm of the early years started to wear off, an effect that is, perhaps, inevitable in any line of research whose early promise of boundless validity and applicability finally has to give way to a realistic acknowledgment of the boundaries of its concepts and the limitations of its measures. But work on implicit motives also came under fire from critics (e.g., Entwisle, 1972; Fineman, 1977; Klinger, 1971), and the leading researchers of the field took an amazingly long time to muster convincing rebuttals to the inappropriate criticisms and to learn from the appropriate ones.

It is ironic and also tragic that the very principle on which implicit motive research is built—that people cannot validly report on their motivational needs, and, therefore, indirect assessment methods have to be employed—was ignored by many researchers at the fringes of the field, some of whom then complained that measures of implicit motives did not correlate with self-report measures and did not predict the same outcomes as these self-report measures (e.g., Entwisle, 1972; Fineman, 1977). Others simply developed self-report measures for the motivational needs studied by implicit motive researchers and declared these new assessment tools to be able to tap the same constructs as the more time-consuming picture story content-coding motive measures. As a consequence, scores of questionnaire measures of the “achievement motive” and other motivational needs have been developed and established under the same construct name as measures of implicit needs, and entire theories have been built on findings obtained with them, despite the fact that these questionnaire measures do not substantially correlate with implicit motive measures. A novice entering the field of human motivation today is faced with the daunting task of figuring out that self-report and indirect measures of presumably the same motive do not measure the same thing or predict the same kinds of outcomes, despite having the same construct name. McClelland, Koestner, and Weinberger (1989) later commented on this state of affairs: “Another way to react to this lack of correlation is to take it seriously, to

insist that at a minimum, psychologists should not call by the same name two measures that do not correlate with one another [. . .].” (p. 691).

To some extent, however, this state of affairs was also a homemade problem, reflecting a lack of careful theorizing in the field of implicit motives itself, theorizing that was commensurate with basic assumptions and research findings. True, Atkinson (1957) had developed the influential risk-taking theory of achievement motivation. But in doing so, he abandoned a basic assumption of his earlier work—that motivational needs operate unconsciously—and integrated a self-report measure of fear of failure into his research and theorizing, thereby perhaps giving others the impression that questionnaires can be used, after all, to validly assess motivational needs. Atkinson’s later work on the dynamics of action, although developed in part to account for the lacking internal consistency of implicit motive measures, ranged far beyond the domain of the empirical data—too far, perhaps, to provide a useful conceptual framework for predicting, testing, and interpreting effects of implicit motives empirically. McClelland, on the other hand, had witnessed the rise and fall of Clark Hull’s general theory of behavior while he was a graduate student at Yale University and therefore was ambivalent about the usefulness of theories: “My *primary* loyalty is to the phenomenon, to the empirical fact—and if it messes up somebody’s theory, so much the worse for the theory” (McClelland, 1984, p. 28, italics in original).

What was missing at the very core of implicit motive research was an explicit recognition that it is not sufficient to postulate that people have no insight into their motivational needs and then prove the point by showing that self-report measures of motivation do not correlate with implicit motive measures. That only leads to the question: If they are not correlated, which is better at predicting relevant criteria? The issue of statistical independence then becomes an issue of competing validity, and only one type of construct can win the race. What needs to be added to the independence of the predictors is, at a minimum, a conceptual and empirical specification of which kinds of outcomes are affected by each type of measure.

A study by deCharms, Morrison, Reitman, and McClelland (1955) provided the clues necessary to arrive at such a model. The researchers assessed participants’ implicit achievement motive and also the value they explicitly placed on achievement. Then they had them work on a classic validity correlate of the implicit motive (an anagram task) and on a task that required them to ascribe achievement-related traits to a target person. The implicit achievement motive predicted performance on the anagram task, but not the traits participants ascribed to the target. In contrast, the degree to which participants rated achievement as something valuable predicted their trait ascriptions, but not their anagram task performance. What this study suggested, then, was that implicit motivational needs predict behavioral performance and that self-attributed motives predict verbal choices and attitudes.

Either this set of findings was so perfectly compatible with McClelland's and his colleagues' way of thinking about conscious and unconscious levels of motivation that they did not fully realize its significance for a person unfamiliar with their views on motivation (i.e., someone outside of the field of implicit motives), or they failed to grasp that they had just identified a rift between levels of behavior that was much more general and pervasive than the mere dissociation of the predictor measures of implicit and self-attributed motivation. Whatever the reason, McClelland and his associates never made much of these findings for the next 25 years.

But in light of waxing criticism of implicit motive measures and waning interest in implicit motive research, McClelland finally realized that something more was needed than an insistence on the validity of the implicit motive measure. After all, as long as implicit and self-attributed motives were in competition with each other and one was presented as a valid measure, where, exactly, did that leave the other?

McClelland (1980) provided the first grudging acknowledgement that self-reports of motivation might also have some validity, but in a different domain of behavior than implicit motives. Using data from studies on academic and life success, early socialization, and other validity correlates, he showed that implicit motives and self-attributed needs (which he termed "values") predicted different kinds of behavior: implicit motives, long-term behavioral trends in unstructured situations; and self-attributed motives, short-term behavior on questionnaires and in response to specific, clear-cut situational demands. At the same time, he questioned the evidence proponents of self-report measures of motivation cited for the reliability and validity of their constructs. In particular, he denied that self-report measures of motivation assessed anything motivational at all. Although many of his arguments against self-report measures of motivation were incisive and still deserve careful consideration and a principled response by anyone who considers using self-report measures of motivation, this half-hearted concession did not yet resolve the antagonism that the field of implicit motive research had developed with the rest of personality psychology, then a notoriously self-report-prone field of research.

Conceptual Refinement and Reinvigoration: 1989 until Now

During the 1980s, McClelland kept working on a way to integrate conscious and nonconscious forms of motivation into a coherent account of motivated behavior (McClelland, 1985, 1987). Meanwhile, the rest of psychology caught up with implicit motive research. Cognitive psychology, which for a long time had put its trust in common-coding models of information processing, started to realize that information that is processed consciously takes a different route and has different properties than information that is processed outside of

consciousness (e.g., Squire, 1986). Concepts like implicit perception, implicit memory, and implicit learning started to become popular, and the paradigms to test and separate them from explicit forms of cognition were developed (Kihlstrom, 1990). Social psychologists began to realize that attitudes can be processed and represented implicitly, outside of conscious awareness, and can dissociate from explicit, conscious attitudes (e.g., Devine, 1989). Neuropsychologists had known at least since the 1950s that the brain has several dissociable systems for perceiving, processing, and storing information, only some of which are involved in conscious memory and control of behavior (e.g., Corkin, 1968). In short, psychology collectively realized that the human psyche operates on more than one level, and that this can be robustly demonstrated across many different mental functions (Kihlstrom, 1990).

So the time was ripe when in 1989, McClelland, Koestner, and Weinberger published a seminal paper, "How do self-attributed and implicit motives differ?," in *Psychological Review*. In this article, they proposed that implicit motives respond to task-intrinsic incentives and influence operant behaviors (i.e., behavior in unstructured situations), whereas self-attributed (or explicit) motives respond to social incentives and influence respondent behaviors (i.e., behaviors in response to a specific social demand or expectation). Besides formulating for the first time a comprehensive, conceptually symmetric model for conscious and nonconscious levels of motivational control over behavior, McClelland et al. also introduced new terms for existing measures and constructs that have been adopted by most researchers in this field since then. For one, they coined the term *implicit motive* in analogy to Schachter's (1987) implicit memory concept to denote nonconscious motivational needs assessed through indirect means and contrasted it with the term *explicit motive*, which denoted the motivational needs and strivings that people consciously ascribe to themselves. A second change in terminology was the switch from TAT to Picture Story Exercise (PSE) as the official name for the picture story methods commonly used to assess implicit motives. The change reflects the fact that motive researchers have rarely used Murray's (1943) original TAT stimuli or the administration procedures associated with it. Instead, they frequently used pictures coming from other sources, administered the test in group settings in which participants would write their stories under timed conditions instead of having them tell their stories to an interviewer, and they used empirically derived content-coding systems for scoring the stories instead of coding systems developed based on psychodynamic theories or clinicians' consensus (Winter, 1999).

A meta-analysis by Spangler (1992) soon corroborated the fundamental validity of McClelland et al.'s (1989) two-systems model. Spangler found that across hundreds of studies, the implicit achievement motive was indeed a powerful predictor of operant behaviors, particularly if suitable task incentives were present, but not of respondent behaviors, and that the reverse was true of the explicit achievement motive. These findings and McClelland et al.'s (1989)

detailed model of motivation went a long way toward reconciling implicit motive research with the mainstream of personality psychology and convincing a new generation of researchers that the implicit motive construct was worth investigating.

This development was further facilitated by the publication of a second edited volume dedicated to the theory and assessment of implicit motives and related constructs, Smith's (1992) *Motivation And Personality: Handbook of Thematic Content Analysis*. The book featured, besides new and reprinted chapters on conceptual issues related to implicit motive measurement and validity, coding manuals for classic, revised, and new measures of implicit motives as well as cognitive styles that are expressed in people's storytelling. It also contains chapters summarizing the validation of each coding system and, in the appendix, training materials for the acquisition of coding skills in all of the featured systems. Together with Atkinson (1958), this book represents an authoritative and comprehensive source for the reader who is interested in learning how to code text for motivational imagery.

Perhaps a third, albeit less visible, promoter of the rejuvenated interest in implicit motives was the fact that since the 1970s, David Winter had developed an integrated coding system for the assessment of motivational imagery in political speeches and historical documents (Winter, 1991). This running text system allows researchers to code achievement, power, and affiliation imagery in one run, using simplified coding rules. Although developed for other purposes than for use with the PSE, many researchers have started using Winter's running text system for coding PSE stories due to its efficiency and comprehensiveness.

So after a time of stagnation, research on implicit motives was off to a fresh start, benefitting from the fact that the zeitgeist in academic psychology had finally caught up with its basic assumptions, from new and better models, and a greater and more refined choice of coding systems. A factor that has also started to play into the rapidly increasing sophistication of implicit motive research is the fact that cognitive psychology has developed models and instruments that now help motive researchers tease apart the domains and boundaries of influence of implicit and explicit motives on a variety of processes, such as attention, learning, and memory. And biopsychological approaches to motivation, with their conceptual rigor and precision, hold particular promise for understanding the neurobiological foundations of implicit motives and the behaviors affected by them (Berridge, 2004; LeDoux, 2002). So perhaps it is not too much to hope for if we predict that after decades of relative separation from the mainstream of psychological research, the implicit motive construct will become a useful ingredient of affective neuroscience approaches to explaining behavior, one of the most rapidly growing disciplines of psychology.

We will not go into much detail about the specific developments in motive research after 1989; these are described at length in the many excellent chapters contained in this book. Instead, as a measure of the resurgence of

interest in implicit motive research, we would like to note that implicit motives are featured at length in several popular textbooks of personality psychology (Carver & Scheier, 2007; Larsen & Buss, 2008; McAdams, 2009; Winter, 1996), many of which are enjoying widespread use in undergraduate courses and are frequently updated in new editions. The *Handbook of Personality* (John, Robins, & Pervin, 2008), a defining resource for the field of personality psychology and the training of graduate students, now features in its third edition a chapter on implicit motives for the first time (Schultheiss, 2008). Its companion *Handbook of Research Methods in Personality Psychology* (Robins, Fraley, & Krueger, 2007) contains a chapter dedicated to implicit motive assessment (Schultheiss & Pang, 2007). Thus, the implicit motive construct enjoys continued as well as newfound popularity among personality psychologists and also in other disciplines, and we are confident that its impact will continue to increase as more researchers start using in their own work the conceptual and methodological tools developed by motive researchers. The purpose of this book is to document the more recent developments in implicit motive research and to be a resource for colleagues who would like to familiarize themselves with established and emergent theories and measurement approaches in the field.

Common Principles of Implicit Motive Research

Research in the field of implicit motives is defined by a set of common principles and assumptions about what implicit motives are, how they can be assessed, and how they operate. We have already mentioned some of them above; nevertheless, we think it is helpful to be absolutely clear about the assumptions guiding the study of implicit motives because they often are not made fully explicit in the work. Assumptions are fundamental, broad hypotheses underlying the actual, narrow hypotheses and the way they are tested. As such, they might be wrong (slightly or wholly) and not only for the sake of falsifiability, but also for rational and principled development of models and theories, they need to be spelled out. So here are, in brief, the common principles guiding most of the work presented by the contributors to this volume:

1. Implicit Motives Are Nonconscious and Cannot Be Measured Through Self-Report

As we have pointed out previously, this assumption has guided the field from the very beginning and has also been supported in numerous studies that have examined the correlation between the motivational needs that individuals attribute to themselves and PSE measures of the same motivational needs. Across

studies and motive domains, the correlation is close to 0. This is even the case when the explicit measure is made as similar as possible to the implicit measure, as Schultheiss, Yankova, Dirlikov, and Schad (2009) have recently demonstrated.

2. Situational Arousal of a Motivational Need Is Associated with Characteristic Changes in Thought Content and Other Nondeclarative Markers of Motivation

This was Atkinson and McClelland's (1948) fundamental insight and idea: We may not know how a motive is manifested in behavior, but we can find a way to arouse it and then examine how thought content as manifested in picture stories changes as a function of the arousal. Virtually all motive measures were derived following this basic principle or at least validated through their convergence with measures developed in this way. There can be considerable variation in the type of situational arousal, ranging from food deprivation (Atkinson & McClelland, 1948) to subliminal tachistoscopic priming with brief sentences (Siegel & Weinberger, 1998). Although most studies so far have used changes in imaginative stories written after the situational arousal had taken place, and thought content may be a good place to start looking for arousal-induced changes, there is no a priori reason why this approach could not also be extended to other markers of aroused motivational states.

3. Motives Represent Capacities for Specific Affective Experiences; They Orient, Select, and Energize Behavior

Starting with McClelland et al.'s (1953) work, motive researchers have conceived of motives as dispositions to seek out certain incentives for the affective changes they elicit. Because a person with a strong motive is a person who has a strong affective response to an incentive, the person orients attention toward cues predicting the possibility of such an affective experience, selects through learning predictive cues and instrumental behaviors that will allow approach toward and attainment of the incentive, and executes such behaviors with increased vigor and energy. These properties represent the hallmarks of motivation in studies with animals and humans alike (Berridge, 2004; McClelland, 1987; Schultheiss, 2008).

4. Motives Interact with Situational Incentives to Shape Behavior

This interactional view of motivation is part and parcel of classic and modern theories of motivation (e.g., Lewin, 1935; Toates, 1986) and has been a

fundamental assumption of implicit motive research from the very beginning. For example, in the domain of achievement motivation, one and the same situational incentive (e.g., a task of moderate difficulty) may elicit dramatically different responses (e.g., an increase or a decrease in effort expenditure) depending on the strength of an individual's implicit motive to achieve. Conversely, the behavioral effects of implicit motives can markedly differ when critical situational features are varied in laboratory settings or real-life situations. For instance, the implicit achievement motive predicts response latencies on a mental concentration task when participants receive feedback that their performance deteriorates but not when the feedback indicates that they are doing well (Brunstein & Maier, 2005). Motive researchers generally adhere to Lewin's (1935) view that behavior might best be understood as the product of person and situation variables and thus seek to elucidate how implicit motives interact with environmental cues to generate meaningful patterns of behavior both within and across situations.

5. Motives Have Pervasive Effects Across Several Levels of Psychological Functioning

Because early researchers—McClelland in particular—kept pushing the envelope with regard to how far the influence of implicit motive reaches, we now know that motives can have detectable effects at many levels: in the biological basement of brain and body; at the first-floor level of individuals' cognitive, affective, and behavioral functioning; and all the way up to the attic of societal, historic, and economic phenomena—a pretty breathtaking span of validity for a construct! But perhaps this span is not surprising because motives can be viewed as individual manifestations and elaborations of fundamental systems that have guided the behavior of our species in phylogenetic timespans (including historic time) and in many different environments and situations. They are bound to have marked effects on behavior at all levels.

6. There Is a Limited Number of Implicit Motives

Power, affiliation, and achievement, the motives this volume focuses on, are not the only basic motivational needs, and others, such as hunger or sex, also deserve more attention in research. However, the list of potential motives is not endless and in all likelihood limited to the small number of phylogenetically evolved motivational systems biopsychologists have described in some detail now (see, for instance, Panksepp, 1998).

Overview of the Book

The book is divided into four parts. The first part, titled “Motive Systems,” provides an overview of past and current research and theorizing on the “Big Three” of motivation research, the needs for power, achievement, and affiliation. In Chapter 1, Fodor portrays the power motive as an ambivalent force behind human behavior: sometimes beneficial, such as when power motivation fuels creativity; sometimes detrimental, such as when power-motivated individuals become stressed by dominance challenges or elicit ingratiating behavior in members of their work teams. Pang, in Chapter 2, chronicles the metamorphoses of the achievement motive, from its beginnings in *The Achievement Motive* (McClelland et al., 1953) and Atkinson’s (1957) influential risk-taking theory, to the work of Heinz Heckhausen (1963) in Germany who dissected the motive into a hope of success and a fear of failure component, all the way to the latest rigorous research efforts to identify the correlates of achievement hope and fear in the thoughts and behaviors of research participants. Weinberger, Cotler, and Fishman chronicle and discuss the measurement and validity of the affiliation motive in Chapter 3. Like the achievement motive, the affiliation motive consists of hope and fear components with different validity correlates: a capacity for closeness and love, also called intimacy motivation, and a fear of rejection and loneliness, captured to a large extent in classical measures of affiliation motivation. As Weinberger, Cotler, and Fishman point out, this distinction deserves greater attention in future measurement and theorizing about the motivational need to affiliate. Langens’s chapter on activity inhibition concludes the section on motive systems. Activity inhibition does not represent an implicit motive in its own right, but it has been identified as an important implicit measure of self-regulation that frequently influences the expression of motives in behavior. Because it was derived atheoretically, as a lucky finding in one of the first attempts to aid language analysis through the use of the computer in the 1960s, its impact on the behavioral expression of power and affiliation motivation was long regarded as a slight embarrassment by scholars in the field. The situation has rapidly changed in recent years, however, because researchers have started to study activity inhibition in its own right and have gained important insights into the properties and functions of this construct. The fascinating findings from this research, to which Langens has made key contributions, are summarized in Chapter 4.

The second section of this book is titled “Assessment of Implicit Motives,” and the chapters in it deal with various overarching issues in motive assessment. In Chapter 5, Pang gives a state-of-the-art account of the craft of motive assessment based on content coding of picture stories. Her contribution provides excellent guidance for all the major decisions that have to be made and issues that need to be considered if a researcher wants to employ implicit motive

measures: which and how many pictures to select, whether to pretest pictures, which coding systems to use, how to train coders, how to ensure high interrater reliability, and much more. Her chapter features descriptive information about the pull of a large array of frequently used pictures for the motivational needs for power, achievement, and affiliation—information that will help make the compilation of suitable picture sets for the assessment of a given motive or sets of motives less of an art and more of an exact science. Pang also sketches out her own work on a new coding system for *n* Achievement, which separates hope of success and fear of failure based on sophisticated motivational arousal experiments.

Brunstein and Schmitt (Chapter 6) present the exciting results of their efforts to develop and validate an alternative approach to motive assessment based on the Implicit Association Test (IAT; Greenwald, McGhee, & Schwartz, 1998), an approach that promises a highly overdue extension of the measurement basis of implicit motives. As they demonstrate in considerable detail, their IAT measure of *n* Achievement shows convergent validity with a classic PSE measure of *n* Achievement, predicts the same criterion (physiological indicators of effort investment of easy, medium, and difficult tasks), and fails to overlap with self-report measures of achievement motivation.

Finally, Blankenship discusses the use of the computer in motivational theory and assessment in Chapter 7. In the past, computers have been used for the development and validation of a sophisticated theory of motivational processes, Atkinson and Birch's (1970) dynamics of action theory, and Blankenship highlights the basic assumptions behind this theory and its application in computer simulations that illustrate how stable motives can give rise to variable behavior. She also discusses how the computer can be used in the assessment of motives and in the analysis of thematic content in the stories collected from research participants.

By being the longest of the book, the third section titled "Basic Concepts and Processes" reflects the fact that the field of implicit motive research has been reinvigorated in recent years by a broad and rigorous exploration of the conceptual foundations and implications of the implicit motive construct. In Chapter 8, Bender and Woike summarize others' and their own impressive work on the effects of implicit motives on memory processes. Motives enhance memory for events and episodes of daily life, particularly when these episodes are affectively charged. Integrating motivational and cognitive models of memory, Bender and Woike portray the effects of motives on each stage of information processing contributing to this robust and pervasive effect.

In Chapter 9, Stanton, Hall, and Schultheiss discuss the nature of motive-specific incentives, whose properties have not been identified and described satisfactorily for a long time. They suggest that implicit motives respond preferentially to nonverbal incentive stimuli and contrast this mode of functioning with explicit motives, which respond primarily to verbal incentives. They also

present a more specific model derived from the nonverbal-processing hypothesis, motivational field theory, which holds that nonverbal emotional signals of a sender serve as motivational incentives for the perceiver, an effect that depends critically on the perceiver's implicit motives.

Hall, Stanton, and Schultheiss then descend into the biological basement of motives in Chapter 10. As these authors point out, evidence for a strong and specific biological basis for each of the major three implicit motives has been accumulating since the 1960s, and this evidence may be among the strongest for the validity of the implicit motive construct because it gets very close to the actual neurophysiological substrates of motivation. Hall, Stanton, and Schultheiss provide a review of the older literature on the biopsychology of motives and integrate it with more recent work on the behavioral endocrinology of the big three motives. They also report findings from recent brain imaging work on the role of the needs for power, affiliation, and achievement on brain activation responses to facial expressions of emotion.

The next two chapters of this section deal with a fundamental and pervasive finding in motive research and its implications: the statistical independence between implicit motives and the motivational needs and strivings people ascribe to themselves on questionnaires. Thrash, Cassidy, Maruskin, and Elliot (Chapter 11) provide a careful and thought-stimulating analysis of the methodological and conceptual reasons for the low overlap between implicit and explicit motives. They also discuss a variety of moderators that influence the degree to which people's implicit motives and self-attributed needs converge and that have emerged in research conducted over the past 10 years. In Chapter 12, Brunstein then discusses the consequences of the matches and mismatches between implicit motives and explicit goal strivings for emotional well-being. His review of the literature clearly shows that although implicit motives and explicit goals represent independently operating systems for the regulation of behavior, they interact in shaping feelings of elation and dejection that depend on the degree to which individuals succeed or fail at motive-congruent goals. In tandem, the chapters by Thrash et al. and Brunstein emphasize the validity of the basic assumption of implicit motive research—that people generally do not have introspective access to their motivational needs—and demonstrate how the concepts and methods developed by the field can be used for addressing one of the fundamental issues of psychology since Freud, the independence of and dynamic interactions between conscious and unconscious realms of the psychic apparatus.

Baumann, Kazén, and Kuhl (Chapter 13) present a principled conceptual account of implicit motives from the perspective of personality systems interaction (PSI) theory. PSI theory holds that affective and cognitive macrosystems interact in shaping people's volitional, motivational, and self-regulatory adjustment to situational demands and affordances. According to Baumann, Kazén, and Kuhl, implicit motives are rooted in one of the cognitive macrosystems,

extension memory, and moderate the interactions between and typical configurations of affective and cognitive systems. PSI theory also yields a new, more differentiated measurement approach for the assessment of each of the big three motives, the Operant Motive Test, whose measurement credentials the authors describe and discuss in this chapter.

The fourth and final section of the book deals with “Interdisciplinary and Applied Aspects” of implicit motives. Winter (Chapter 14) discusses the political and historical correlates of implicit motives and provides a breathtaking vista of the findings he and his collaborators have collected over the years. His research shows that the motivational needs of political leaders predict their behavior in office and that the secret to belligerent or peaceful resolutions of conflicts between countries is the degree to which the communication between the parties involved reflects power-related or affiliative themes. Here is hope that some fundamental laws driving the course of history can be identified and measured empirically and that this knowledge may one day be used to predict and prevent violent conflicts.

Since McClelland’s (1961) seminal work on the cultural origins and correlates of the achievement motive, one of the strengths of research of implicit motives has been its focus on the cross-cultural commonalities of and differences between the eliciting conditions and behavioral manifestations of motivational needs. Like few others in recent years, Hofer has pushed the limits on the effects of culture on motives with his rigorous and painstaking work on cultural influences on motives and their effects on well-being and behavior in Germany, Costa Rica, Zambia, and other countries. He provides an overview of the key findings resulting from this impressive research program and integrates them with earlier work in Chapter 15.

In contrast to the long tradition of cross-cultural work on motives, their role in clinically relevant syndromes and disorders has received scarce attention in research. This is particularly amazing in light of the fact that implicit motives play a critical role in people’s emotional well-being (see Chapter 12), and it therefore appears likely that they also might influence the occurrence of mood disorders. Weinberger, Cotler, and Fishman (Chapter 16) are the first to explore the role of implicit motives as vulnerabilities for and protective factors against psychological problems. Their work on oneness motivation, a member of the family of affiliative needs, indicates that this disposition is associated with beneficial outcomes of psychotherapy and may promote mental health.

Boyatzis and Kelner, in Chapter 17, discuss the role of competencies, which are rooted in, and are measured through similar content-coding procedures as, implicit motives in business and management contexts. Competency assessment has been used successfully to identify individuals who will show superior performance in specific jobs. Boyatzis and Kelner provide an overview of the links between job competencies and other measures of managerial success and job-related emotional intelligence.

Last but not least, Rheinberg and Engeser summarize their work on motivational training of teachers, students, and their parents in Chapter 18. In groundbreaking studies conducted over the past 30 years, Rheinberg has shown that motivational training can help reduce fear of failure in the classroom and bolster hope of success through the setting of individual norms instead of social norms. Rheinberg and Engeser also discuss the issue of motivational competence, that is, individuals' ability to accurately perceive the strength of their implicit motives and to set their goals accordingly. Many of Rheinberg's findings are presented in English language here for the first time.

Science is a collaborative effort, and good scientific concepts need the nurture and care of many brilliant and creative minds. This is also true for research on implicit motives in general and the path that has led us to the conception and publication of this volume in particular. We thank our excellent, knowledgeable contributors for bearing with us through the various stages of this book. Without them and the work that they present here the field would be in a state of intellectual anemia; thanks to them, it is brimming with exciting new ideas, theories, and measures. Thanks to them, we have every reason to hope for future growth and groundbreaking new insights in the field of implicit motives!

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Oliver C. Schultheiss

Joachim C. Brunstein

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SECTION 1

Motive Systems

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Chapter 1

Power Motivation

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Introduction

Need for power (*n* Power) is conceptualized as a desire to influence, control, or impress others and, as a corollary, to receive acclaim or at least recognition for these power-motivated behaviors. As a motive, it both energizes behavior and channels it toward objectives that fulfill the basic need (Carver & Scheier, 2008). The operational definition of *n* Power most in use arose with the coding system developed by David Winter (1973, 1992) in relation to what researchers now term the Picture Story Exercise (PSE; McClelland, Koestner, & Weinberger, 1989; Winter, 1999). McClelland et al. conceived the PSE as a measure of *implicit motives*—that is, motives not readily accessible to the conscious mind but nevertheless capable of shaping people’s feelings and overt behavior in important ways. The term “implicit” seemed appropriate because evidence for the motive is inferred from what people write in response to picture cues. Unlike the self-report inventories that personality psychologists more commonly use, people are not *explicitly* describing themselves. The present chapter explores the *n* Power concept from its origins through the various research applications that it has spawned. Influenced in large degree by Robert MacCleod (1975), one of my professors in graduate school, I believe that we can best understand a psychological concept by tracing it back to its historical origins. For that reason, much of the chapter does exactly that and then shows how current research interests arise in logical fashion from those intellectual roots. Special attention centers on a few topics and issues that have most caught my attention over the years. These include concern about the

reliability of the PSE, the important distinction between implicit and explicit motives, and the concept of power stress.

Measuring Power Motivation

For research-oriented psychologists, the essence of a theoretical concept is its operational definition. As already mentioned, the operational definition of the power motive is the PSE. Presented with individual pictures, the research participant taking the PSE writes 5-minute stories that essentially answer the following questions:

1. What is happening? Who are the people?
2. What has led up to this situation? That is, what has happened in the past?
3. What is being thought? What is wanted? By whom?
4. What will happen? What will be done?

Written instructions exhort participants to write *vivid, imaginative, dramatic* stories, the pictures serving only as a rough guide and not to be viewed as constraining.

An Illustrative Example

Here is a story that is high in power imagery, written by a student who obtained the highest overall *n* Power score in a recent study. The story illustrates some of the scoring categories (italicized phrases) in Winter's scoring system. The picture shows a man and a woman performing a trapeze act.

The people are *Superwoman* and *her arch nemesis Rhinosorous (sic) Man*. She is finally going to catch him and rid the world of his evil ways. Up to this point, Rhino Man has been *terrorizing the world in his attempt to gain ultimate control. He wants to make every person his servant. Superwoman wants to rid the world of his evil ways. The only way she can do this is to knock him into the next galaxy. The people of the world will be forever grateful for her services. For her efforts she will be made Honorary Princess of the World.* She can travel anywhere and do anything she wants all over the world.

Each coding category evidenced in the story receives a score of 1 but does not receive an additional score if it recurs again throughout the story. The first two sentences establish the theme of the story as one of power imagery (coded as Pow Im) and signal that the coder proceed in search of additional categories. If no power imagery occurs, the story receives a score of 0. Power motivation involves concerns about status,

either positive or negative, so the name designation *Superwoman* implies positive prestige of a key actor (Pa+) and the designation *Rhinosorous Man* later described as terrorizing the world suggests negative prestige of a key actor (Pa-). Reference to the recognition that comes with status constitutes power imagery, the reasoning goes, regardless of whether status is positive or negative. Terrorizing the world qualifies as instrumental activity oriented toward achieving the power goal (I). Wanting to make every person his servant expresses Rhinosorous Man's need to achieve control over others (N). Wanting to rid the world of Rhinosorous Man's evil ways again expresses a need for control (N), now for Superwoman, but does not receive an additional score. In contemplating how to achieve this goal, Superwoman is showing instrumental activity (I) insofar as she is planning a means to exert power, but because instrumental activity already occurs in the person of Rhinosorous Man, the student writing the story does not receive an additional +1 score. The statement of gratitude from the world at large signifies that Superwoman has produced a momentous and far-reaching effect (Eff) and, like Pa+ and Pa-, indicates thoughts of prestige and recognition. Becoming *Honorary Princess of the World* can be seen either as a further elaboration of Eff or as Pa+. There are additional scoring categories that do not occur in the foregoing story, but the ones mentioned give an idea of how the system works. Total score for this story tallies out at 6. Rarely does a single story command a score higher than 6, although a total score of 11 is theoretically possible.

The remaining five coding categories are as follows:

- Ga+ (anticipation of success in pursuit of the power goal)
- Ga- (anticipation of failure to attain the power goal)
- G+ (positive affect associated with success at achieving the power goal)
- G- (negative affect associated with failure to achieve the power goal)
- Bw (a block in the external world outside the actor that represents an at least temporary barrier to power goal attainment).

The difference between the Ga's and the G's is that the former occur *prior to* culmination of the power-motivated act and the latter subsequent to its completion. Ga- and G-, although they signal anticipated or actual failure in the quest for power, nevertheless factor into the total score because they suggest that the individual is thinking about the exercise of power.

Winter (1973, p. 250) devised a schematic diagram (Fig. 1.1) to illustrate the behavioral sequence reflected in the *n* Power coding system. He conceptualizes the motivational process as consisting of four distinguishable parts: (1) wishes, anticipations, and affective states, along with status attributions (Pa+ and Pa-) likely to accompany these states of mind, (2) possible blockage to expression of the power need, (3) instrumental activity directed toward the goal of achieving

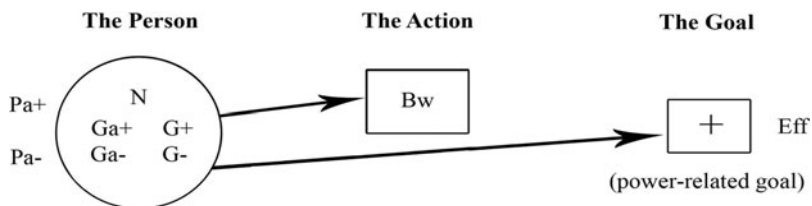


Figure 1.1 Schematic diagram of the *n* Power coding system.

impact, which on occasions may circumvent Bw, and (4) attainment of the power goal, perhaps resulting in the desired Eff.

The foregoing story combines both the beneficent and malevolent inclinations toward which the power motive may impel people. Reflecting on the question of whether power motivation is a good or bad influence on the human condition, Winter (1973) said it well:

To me, power is like *fire*: it can do useful things; it can be fun to play with and to watch; but it must be constantly guarded and trimmed back, lest it burn and destroy. (p. xviii)

Genesis of the Winter Scoring System

Building upon Veroff (1957) and Ullman's (1972) insights on *n* Power and *n* Influence, respectively, and in an attempt to integrate these two formulations with the one he began in his doctoral dissertation (Winter, 1967), Winter (1973) proceeded to develop the scoring system that has guided most of the more recent research on power motivation. The procedure Winter chose was both conceptual and empirical. He showed a group of MBA students a film of the 1961 inauguration and speech of President John F. Kennedy. The film conceptually captured key elements of what we mean in common parlance by the term power, notably, the personal charisma of John Kennedy himself and the grandeur of the occasion (the transfer of elected executive power). A control group of MBA students observed a film in which a businessman discussed science demonstration equipment. Immediately after seeing the film, each group took a six-story version of the PSE. They also completed some questionnaires. The group who saw the Kennedy film endorsed such feelings as pride, leadership, identification with a cause, respect, and fascination. President Kennedy appeared to affect them as a heroic, powerful figure. The control group reported essentially neutral feelings without power connotations.

What Winter then did was empirically search for scoring categories that differentiated between stories written by those MBA students who saw the Kennedy film versus those who witnessed the demonstration of

scientific equipment. A priori ideas of what was “power” guided his search in large part but were superseded by empirical evidence when that evidence showed disagreement with those preconceived expectations. As a final test of procedural adequacy, Winter applied the emerging scoring system to MBA students from the original sample whose protocols he had set aside and not scored and whose group membership (aroused versus neutral) he did not know (blind scoring). For example, the Eff category (registered impact felt by another) was prevalent among stories written by MBA students who observed the Kennedy video but rare among those exposed to the equipment demonstration. Some research participants who take the PSE cold, that is, in the absence of conditions designed to arouse the power motive, achieve *n* Power scores just as high as most persons subjected to conditions of arousal. That is the way their minds normally connect with the social world around them, much in the same way the MBAs felt prompted to think in response to the Kennedy film.

The empirical soundness of this procedure strikes me as especially compelling in light of an event I experienced during the first experiment I conducted with the Winter scoring system. The research participants were MBA students who took the PSE in small groups. On one occasion, four MBAs gathered shortly prior to my arrival in the designated room for taking the PSE. They had just left their evening class. Although only a few minutes had passed before I walked in, they had become embroiled in a heated, acrimonious debate as to the proper interpretation of a business study case they had examined in class. This was more than a case of intellectual disagreement. I had difficulty calming them. The PSE stories they then wrote vastly exceeded the pre-established standard for qualifying as “high” scorers in the forthcoming laboratory experiment scheduled for a few days later. The statistical probability of these scores all occurring at that level under normal circumstances was very slim. The decision, of course, was to exclude them from the actual experiment.

Early Beginnings

The PSE measure of *n* Power that Winter developed derives in part from the Thematic Apperception Test (TAT) originally devised by Henry Murray (1938). The PSE differs markedly from the TAT, however (Winter, 1999). It is more thoroughly grounded in empirical verification procedures. Trained raters analyze written stories by means of a precise and detailed coding system. By contrast, no empirically derived coding system accompanies the traditional TAT. Individual raters evaluate stories based on their own theoretical perspectives and methods of gaining insight. Hence, there can occur a lack of consistency of interpretation across raters. As a projective test, the PSE is predicated

largely on the assumption that the stories people write express unconscious motivations. Although scholars attribute the seminal notion of the unconscious to Sigmund Freud, it is interesting to note that without any obvious reference to Freud, the novelist Joseph Conrad (1921) appears to have formulated the same concept:

... it is my belief that no man ever understands quite his own artful dodges to escape from the grim shadow of self-knowledge. (p. 64)

Robert Louis Stevenson predated both Freud and Conrad with *Strange Case of Dr. Jekyll and Mr. Hyde*, which was published in 1885 and already contained a rather vivid depiction of the struggle between rational thought and self-control on the one hand and hedonic striving and impulsivity on the other.

Winter (1973) reasoned that the power motive should be associated with occupying positions of office in student organizations in which students can anticipate opportunities to exercise power. He found that office-holders did indeed exhibit high *n* Power scores by comparison with students who did not hold offices in student organizations. He further obtained evidence that college athletes who were varsity letter winners in directly competitive sports (football, basketball, wrestling, hockey, etc.) scored distinctly high by contrast with other students. The man-against-man or team-against-team aspect of these sports in itself provides multiple opportunities for the exercise of power. Indeed, the physical activity that can occur in many of these sports often assumes violent forms. The recognition that comes to varsity athletes of course is well known, as is the realization that this recognition continues beyond the college years. Students who aspired to become teachers also were high in *n* Power. The finding makes obvious sense, as teachers possess abundant opportunity to wield influence over their students and shape in part the kinds of people they become. In sum, *n* Power is related to actions and activities that involve formal social power and personal impact.

Winter and his student associates (Winter, 1973) found that college students high in *n* Power chose as close friends those students who were not well known by other students. The chosen students likely posed minimal threat to those high in *n* Power. They did not compete with them for power and prestige. As persons who were socially "disinherited," they probably were well disposed toward accepting the attention and friendship that the power-motivated students were willing to give them, even if that meant accepting some control over their lives by another person. High-*n*-Power persons, therefore, achieve solidarity within the group and enhance self-esteem in persons of low recognition and status. Power-motivated men, their research further showed, had a preference for

unassertive, dependent wives. Overall, *n* Power seems to involve a concern that one not accept much by way of influence from others.

Reliability Issues

Psychometrically minded psychologists (e.g., Entwisle, 1972) have criticized the PSE on grounds that, unlike self-report personality questionnaires, the PSE lacks test–retest reliability. However, the problem may be an artifact deriving from test instructions, which state that the participant should try to write *imaginative* stories. On retesting, the participant may strive so hard to be creatively different from the first occasion that expression of the true motive becomes distorted. An attempt to show oneself as ideationally versatile may overshadow the actual motives that ordinarily shape the writing of PSE stories. When the instructions on retesting state that the participant may now write stories that are similar to *or* different from the ones written on the previous occasion, test–retest reliabilities rise to the .60s (Lundy, 1985; Winter & Stewart, 1977).

In considering the question of internal consistency, it is important to note the distinction between the usual self-report inventory and the PSE. Self-report inventory items such as those measuring introversion/extraversion are closely similar to one another in meaning. With the PSE, this is not so. The various pictures differ in the amount of “pull” they exert for a given need (Carver & Scheier, 2008). When the pictures are similar in the degree to which they suggest power imagery, Cronbach’s alpha coefficient is acceptably high, .70 in one study (Fodor, Wick, & Hartsen, 2006).

McAdams (2006) emphasizes a fundamental distinction between motives and traits that goes far to explain why PSE measures of motive strength fail to achieve high levels of test–retest reliability. As the McClelland tradition defines them, motives do not hold absolutely steady over time; they vary in strength. Certain conditions intensify them, others fulfill them. Nevertheless, the strength of a given motive stays within a boundary that characterizes an individual, so that an individual’s score at a given time can predict future behavior years later. Traits as measured by self-report remain more consistent even over long periods. Momentary incentives register little effect.

The Distinction Between Implicit and Explicit Motives

The thinking behind use of PSE measures to study motivation is that these measures reflect motivational processes that operate outside a person’s

conscious awareness. For this reason, McClelland, Koestner, and Weinberger (1989) introduced the term *implicit motives* to describe them. Evidence indicates that PSE measures do not measure *explicit motives* as expressed through self-report questionnaires (deCharms, Morrison, Reitman, & McClelland, 1955; Schultheiss & Brunstein, 2001). The distinction between implicit motives and explicit motives (or traits) is an important one and forms the very basis for use of the PSE as a measure of motive strength.

Writing on the distinction between motives and traits, Winter, John, Stewart, Klohnen, and Duncan (1998) suggested that *motive* (as assessed by the PSE) and *trait* (as measured by self-report questionnaires) reflect conceptually different aspects of motivation but interact with one another in channeling the individual's behavior over the life course. They reasoned that motives involve wishes and goals that often are nonconscious, that is, *implicit*. These wishes and goals do not necessarily attain fulfillment through one's life experience. Traits, Winter et al. defined as "comprised of more-or-less consistent, generalized, intercorrelated clusters of behavior" (p. 233). They channel the ways motives are expressed, essentially acting as a rudder that steers one through the vicissitudes that define human experience, sometimes approaching and sometimes avoiding specific goals according to where the individual fits along the trait dimension.

Studying Radcliffe and Mills college graduates, Winter et al. examined the interactive effects of both the affiliation and power motives, each in combination with measures of the trait introversion/extraversion. Neither motive correlated with introversion/extraversion score but in combination with introversion/extraversion score predicted important outcomes at midlife. As predicted, women high in *n* Affiliation who were also high in extraversion became more involved in volunteer work than did women high in *n* Affiliation but low in extraversion (introverts). Taken by themselves, neither *n* Affiliation nor extraversion correlated with volunteer work. For introverts high in the affiliation motive, volunteer work presumably would bring on too much arousal, whereas extraverts high in the same motive would seek out the social stimulation that volunteer work provides. Also as predicted and by the same line of reasoning, extraverts high in *n* Affiliation were more likely to combine work and family roles than were introverts high in the affiliation motive. Balancing family against work commitments entails much arousal which affiliation-motivated extraverts likely would find totally acceptable. Lastly, affiliation-motivated women low on extraversion (introverts) were more likely to have separated, divorced, or experienced stress in their close relationships than were affiliation-motivated women high in extraversion. Affiliation-motivated introverts, the reasoning goes, should find intimate relationships overarousing, conflictual, and threatening given their preferences for privacy and solitude. That is, there is a conflict between need for closeness (*n* Affiliation) and aversion to overarousal (introversion) that is brought on by

close social contact. High *n* Power in combination with extraversion predicted entry into *impact careers* (e.g., teaching, management) for the Radcliffe sample, a finding that makes sense in terms of the channeling hypothesis that Winter et al. put forth. Extraverted women high in power motivation likewise placed greater value on work relationships in their chosen career than did introverted women high on the power motive.

Drinking Behavior

David McClelland and his associates (McClelland, Davis, Kalin, & Wanner, 1972) formulated a theory that high power motivation coupled with a low level of restraining thoughts about power inclines men toward heavy alcoholic beverage consumption. They termed this combination of traits *personal power*. Seligman, Walker, and Rosenhan (2001) describe the known effects of alcohol consumption. Physiologically speaking, alcohol in its short-term effects produces sensations of immediate strength (what some college students at my university aptly term “beer muscles”). Absorbed through the stomach wall directly into the bloodstream, it rapidly increases blood sugar, thereby increasing energy availability. To some degree, the process increases adrenalin flow as well. Vasodilation produces sensations of bodily warmth. The more one drinks (to a certain point), the more the mind becomes suffused with a less inhibited type of thinking about personal dominance, notably, about aggression and exploitative sex.

The theory evolved through a series of studies that the McClelland group conducted in a variety of natural social settings, ostensibly for the purpose of studying the effects of various social situations on fantasy. College undergraduates participated in one of these studies, and MBAs in another. In the college undergraduate study, individual groups consisted of 3–6 members. Half the groups, randomly chosen, drank alcoholic beverages in a party setting, and the other half nonalcoholic drinks. They all wrote PSE stories before the party began, another set to different pictures further into the party, and still another set later on. The research plan included procedures for estimating the amounts of alcohol research participants consumed. PSE analysis disclosed that alcoholic beverage consumption as it progressed did indeed heighten fantasies of personal power (high *n* Power coupled with low inhibition about aggression and exploitative sex). Moreover, the more men showed this combined set of traits in their initial PSE scores, the more alcoholic beverage they consumed. The MBA student study was similar, except that the groups received an invitation to attend a discussion in the living room of an apartment rather than attend a party. Wondering if the results they obtained with college students would apply equally to older men whose drinking habits had become stabilized beyond the youthful exuberance and experimentation that characterize the

college population, McClelland and Davis (1972) conducted research at a "working-class" bar where men habitually drank socially in evenings with other men. They wanted to study a group of men whose habits, values, and attitudes likely differed from college students who attend weekend cocktail parties. An advertisement appeared in the local newspaper, specifically mentioning that the study involved the use of alcohol, the telling of a few imaginative stories, and compensation in the amount of \$15 (a significant sum at the time). Volunteers reported to an employment agency and provided responses to questionnaires including information as to whether they were light, moderate, or heavy drinkers. Persons who were receiving treatment for alcoholism or who were trying to refrain from drinking did not receive an invitation to participate further.

A few days later, volunteers reported to a local bar at 8:00 and stayed until 9:30. After a few minutes of small talk, one among a number of graduate students in clinical psychology presented four appropriately selected PSE pictures and asked individual participants to tell imaginative stories about each picture, which the graduate student thereupon tape recorded. A waitress took orders for drinks. She recorded what and how much a participant drank. There were no restrictions on what or how many drinks they could order. They were free to do whatever they pleased within the barroom setting. That is, they could talk among themselves, watch television, play pinball, or talk with other customers. At 9:30, they told more stories to a second set of four pictures.

Differences in drinking history proved to be a major determinant in behavioral outcome. Specifically, heavy and moderate drinkers at the second PSE administration showed greater increase in personalized power (unrestrained assertiveness, i.e., high *n* Power coupled with low activity inhibition) than did light drinkers. Men with strong power concerns and low levels of restraint in fantasy, the data seemed to indicate, are the men most likely to be heavy drinkers. In summarizing these and related findings, McClelland and Davis saw in the motivational pattern associated with heavy drinking a special type of power imagery:

... a personalized concern in which the world is seen as something of a competitive jungle wherein a man must seek to win out over opponents. Drinking tends to increase generalized power concerns, but particularly personal power concerns which are elevated even more than normal in men with heavy drinking histories. These men, who are already concerned about personal dominance before drinking, are even more concerned after drinking; the inference is that they are using drinking as a means of accentuating their desire to dominate others who would oppose them. They may get direct gratification from thinking about winning out over others in fantasy. (pp. 160–161)

Power Motivation and Ingratiation

Woodrow Wilson was a president whom the psychoanalytically trained political scientists Alexander and Juliette George (1964) characterized as high in power motivation. Woodrow's father, a Presbyterian minister, whom family acquaintances described as harsh in his criticism of young Woodrow, seemed to demand of him standards of deportment and achievement not commonly attained by a boy Woodrow's age. The psychoanalyst Karen Horney (1939) saw in these kinds of circumstances a thwarting of a developing child's need for affection. The consequence, she concluded, is a need for power and prestige, an attempt to demonstrate that one is not to be taken lightly, that one is worthy of notice. Throughout Wilson's career as president of Princeton University, governor of New Jersey, and president of the United States, as well as in his personal life, Wilson showed himself to be especially susceptible to ingratiation, flattery from others that one might expect so intelligent a man as Woodrow Wilson to recognize for its manipulative intent. Colonel Edward House, for example, although he had no formal appointment in the Wilson administration, enjoyed enormous influence with the president. When Wilson delivered an address on a given topic, House would write the president a letter, often extolling the address as the most brilliant statement on that topic ever spoken.

Taking our lead from the George and George biography of Woodrow Wilson, Dana Farrow and I (Fodor & Farrow, 1979) subjected the ingratiation hypothesis to empirical test, using male MBA students as research participants. The format was an industrial simulation experiment in which MBA students who had completed the PSE acted as supervisor in interacting with three supposed workers in the next room who were assembling Tinker Toy models from pictured diagrams, a different model on each of five trials. Communication was via an intercom system. The "workers" were paid actors who had made tape recordings. So as not to interfere with the workers' performance, the experimenter told the student supervisor, communication occurred only at the end of a 5-minute trial. The "supervisor" was to use any and all means at his disposal to motivate the workers toward good performance—exhortations, words of encouragement, threats, and increase or reduction in the amount of pay the workers supposedly received for their participation. The workers, the experimenter further told the MBA student supervisor, were not allowed to respond to the supervisor's comments, lest their response unduly influence the supervisor's ultimate evaluation of them. Otherwise, they were free to make whatever comments to the supervisor they wished, but only at the end of a given trial while waiting for the next to begin. At the end of a trial and hence for each model, the experimenter brought in a card tabulating each worker's

number of completed models. Performance always was roughly average according to the performance norms available to the student supervisor (50th percentile give or take a few percentage points). Worker comments were generally neutral and rather meaningless: for example, “You don’t mind if we smoke do you?”; “Is it okay if we pull down the shades?” At the end of certain trials in the experimental condition, Man C made ingratiating remarks to the student supervisor: “You know, I really like your approach. I can see you’re gonna be a good supervisor”; “Personally, I think you’re doin’ a great job”; then toward the end, “You’ve done pretty good.” Man C’s comments in the control condition were neutral and noningratiating both in content and in emotional tone.

Student supervisors who scored high on the PSE measure of *n* Power (top third of the overall distribution) evaluated Man C’s performance highly when Man C was an ingratiator, more highly than did student supervisors low in power motivation, higher than they evaluated the other workers, and higher than the appraisals high-*n*-Power supervisors gave the noningrating Man C in the control condition. Table 1.1 displays these results. High-*n*-Power supervisors also perceived themselves as having influenced the workers more than did low-*n*-Power supervisors.

These experiments have their humorous moments. I was most eager to meet the gentleman who scored the highest on the PSE *n* Power measure, so I arranged to conduct the postexperimental interview myself rather than consign the task to my research assistant as was our usual procedure. The hour was late, approaching 9:00 p.m. Friday evening. I asked my research assistant if there was anything out of the ordinary in the session that just ended. He said no there was not ... except for the singing. The gentleman sometimes would turn off the intercom during the ongoing work trials so that the workers in the adjoining soundproof room could not hear him, and he would sing in a loud melodious voice that reverberated down the hallway. Interesting, I thought—certainly

Table 1.1 Mean Performance Evaluations Given the Ingratiating Worker

Level of <i>n</i> Power in Supervisor	Performance	
	Ingratiator	Neutral Worker
High		
<i>M</i>	29.3 _a	22.1 _b
<i>SD</i>	4.2	4.8
Low		
<i>M</i>	24.2 _b	23.0 _b
<i>SD</i>	4.9	3.9

Note: Means with differing subscripts differ significantly from each other, *p* < .01.

consistent with the need for recognition aspect of the power motive. I greeted the young man. We shook hands. He was a big guy, built like a football player. I so commented, and yes he had played football as an undergraduate. Nothing else about him seemed out of the ordinary. He was very polite, but he kept looking at his watch. I asked if he had an impending appointment, it being Friday night. He replied that yes he did, that he worked as a bouncer in a local tavern, adding with obvious enthusiasm, "And have I got work to do tonight!"

McAdams, Healy, and Krause (1984) also did an experiment that speaks to the ingratiation theme. They asked college students to describe recent interactions with friends, also listing positive and negative personal characteristics of the people who appeared in these friendship episodes. All participants had previously completed the PSE, which was scored for *n* Intimacy and *n* Power. Their findings with respect to *n* Power strengthen the validity of the *n* Power construct and the PSE measure for assessing it. High-power individuals, unlike low-power individuals, understood friendships as an opportunity to take on a dominant, controlling role. High-intimacy individuals, by contrast, adopted a listening role in their friendship interactions. In citing reasons behind their friendships, they emphasized themes of trust, self-disclosure, and caring for the others' well-being. With its emphasis on display and expansion of the self, power motivation was negatively related to the friendship understandings that high-intimacy individuals expressed.

The McClelland Model

The theoretical framework within which researchers conceive implicit motives derives in large degree from the McClelland model (McClelland, 1958, 1976, 1985; McClelland, Koestner, & Weinberger, 1989). People vary in how they configure and react to events in their social environment. Specifically, they encounter learned cues, or *nonverbal incentives* (see Chapter 6). These nonverbal incentives set into motion motive dispositions in which the individual enjoys special strength, namely, *n* Achievement, *n* Affiliation/Intimacy, or *n* Power. They signal the opportunity to experience emotionally reinforcing activity relative to a given motive. When the individual encounters a nonverbal incentive that aligns well with the individual's capacity to experience an affective state associated with a motive, the outcome is an emotional "kick" (Woiike, 1994), that is, a surge in affect. For the person high in power motivation, the appropriate nonverbal incentive suggests an opportunity to experience feelings of strength, vigor, and energy. As conceived by McClelland, Koestner, and Weinberger (1989), the pleasure the high-*n*-Power person experiences from registering an impact on others results in the hormonal release of

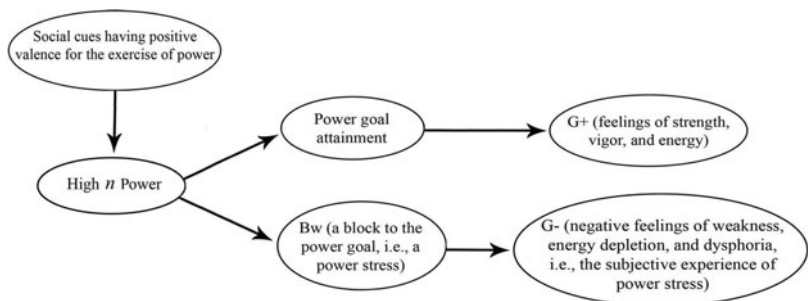


Figure 1.2 Schematic diagram of McClelland's conceptualization of power stress.

norepinephrine, which is a physiological indicator of stress adaptation (cf. McClelland, 1982). Should the power-motivated person not experience the anticipated surge in affect when confronted by an opportunity to exert power, that is, when the power motive is blocked, the result is what McClelland (1976) termed a *power stress*.

The accompanying diagram illustrates the McClelland model. Using Winter's coding symbols, we see the high-power individual reacting to social cues that have positive valence for the exercise of power (i.e., they suggest the opportunity for instrumental activity (I) oriented toward fulfillment of the power need). The need either is successful, thereby resulting in the positive feelings designated as G+; or it is blocked (Bw), eventuating in G-, namely, the negative affect and physiological activation that McClelland defines as power stress. Power stress, in McClelland's analysis, refers both to the thwarting event itself *and* to the reaction to the thwarting event in the high-power person (Fig. 1.2).

Power Stress

McClelland (1976) defined as power stress a social event that simultaneously both arouses and thwarts the power motive. A negative audience reaction to an address by a power-motivated speaker who had intended to create an emotional impact would be an example, as would a thwarted attempt at a committee meeting to persuade others to one's point of view, again if the would-be persuader were high in *n* Power. McClelland saw a similarity between *n* Power and the defining elements researchers had found to characterize persons prone to cardiovascular illness. Most notable among these characteristics were aggression and competitive striving. Supportive of McClelland's view was an experiment by Steele (1973)

that showed especially heightened epinephrine output in the urine as a response to a power-arousal manipulation among research participants whose *n* Power scores increased most as a consequence of the arousal manipulation. An achievement-arousal manipulation did not produce the same effect. Epinephrine production is a key element in Selye's (1973) general adaptation syndrome, which when activated repeatedly and to a high degree is thought to increase the likelihood of cardiovascular illness. Among the effects epinephrine secretion produces are increased heart rate, increased blood pressure, and conversion of glycogen from the liver to blood glucose.

McClelland (1979, 1982) hypothesized that power stress can occur in two possible ways. One way is via an internal control mechanism that bottles up anger, assertiveness, and physiological activation—what McClelland termed *activity inhibition*. The bottling-up process constitutes a power stress in power-motivated persons insofar as it denies expression of the power motive. The second way is via social circumstances that block expression of the power motive or present an insurmountable challenge to its expression. Both methods of incurring power stress—internal restraint or a social event that thwarts the power-motivated person's assertive capabilities—checks any attempt to express power and theoretically can result in harmful physiological consequences if experienced many times.

The mere anticipation that others may resist one's power attempts can cause power stress. Harry Truman was one of the presidents whose inaugural address Winter (1987) coded as distinctly high in need for power. David McCullough (1992) describes how in World War I Captain Truman was assigned the role of commander over 194 men. He a third degree Mason and they a defiant and rowdy bunch of Irish Catholics made the leadership challenge especially intimidating. Previous commanders had been unable to handle them. When he first walked out on the parade field, he felt himself to be completely terror stricken, expecting maximal resistance to his authority. McCullough vividly recounts what is known about Truman's inner feelings at the time. Private Edward McKim recalled that the men observing Harry Truman could plainly see that "he was scared to death." Truman's personal recollections express the stress he felt on that occasion:

Harry said . . . he had never been so terrified. He could feel all their eyes on him, feel them sizing him up. "I could just see my hide on the fence, . . . Never on the front or anywhere else have I been so nervous." (p. 117)

As he was to confess long afterward, writing about himself in the third person, "He was so badly scared he couldn't say a word. . . ." (p. 117)

At last, he called, "Dismissed!" (p. 117)

In researching the power stress concept, McClelland (1979) focused attention on what he termed the *blocked power motive syndrome* (high power motivation, low need for affiliation, high activity inhibition). Affiliation enters into the equation as an antidote to power stress (McClelland, 1985). People high in the affiliation need tend to seek out the company of others as a means of alleviating stress, talking with them, sharing with them their fears and anxieties. McClelland looked at PSE protocols on record 20 years prior to the time he tested men's blood pressure and found a significant correlation between the blocked power motive syndrome and subsequent high blood pressure. Evidence of the syndrome early in life appears to bear implications for cardiovascular health many years later. Studying a male prison population, McClelland, Davidson, Floor, and Saron (1980) sought to determine whether epinephrine was indeed a response to power stress. Persons with the syndrome who also reported high recent levels of power stress did show high epinephrine concentrations in the urine by contrast with appropriate comparison groups.

Of the two ways that McClelland (1979) saw power stress as likely to occur, he conceded that evidence for the influence of thwarting social events was sparse. To round out the evidence, he expressed a need for experiments that present "strong situational challenges" (p. 189) to the power motive. Following through on McClelland's suggestion, I conducted some industrial simulation experiments with male college students. Situational challenges approximated circumstances that managers who participate in training programs commonly report, namely, hard-to-manage work groups (Fodor, 1984) and seemingly irreconcilable conflict among subordinate managers (Fodor, 1985). In the first experiment, the power stress manipulation consisted of preprogrammed comments from supposed workers in the next room. (Actually the workers were fictitious, the comments playacted for a tape recording.) The comments connoted tension within the work group. The "workers" voiced concern about not meeting the standard, about beating out the competition, about not earning enough money, and fear that they not do well enough to be invited back for an additional experiment. For the power stress manipulation in the second experiment, the "company president" of a hypothetical electronics firm supervised two "managers" each with opposing role scripts, one script exhorting a manager to argue for and the other against the manufacture and marketing of a controversial sun lamp. The experimenter informed the "company president" at the very outset of the experiment that it was his task to resolve any possible conflict among his two managers, guiding the group toward an amicable conclusion, and that his ability to do so was an index of his managerial potential. A debriefing occurred at the end of the session. In both experiments, control groups confronted an

absence of power stress. In the first experiment, the “workers” in the control group voiced neutral comments without any hint of tension occurring within the group. In the second experiment, the “managers” in the control group both had role scripts encouraging them to favor manufacture of the sun lamp. High-*n*-Power participants evinced higher stress reactions than low-*n*-Power participants in the power stress condition in both experiments and higher stress than all participants in the control condition. Scores on Thayer’s (1978) Activation–Deactivation Check List comprised the dependent measure in the first experiment and electromyographic recordings from the forearm extensor muscle in the second. Measures of muscle tension are conceived as an index of physiological arousal (Levenson, 1983).

Noting that the hypothalamic-pituitary-adrenal (HPA) axis responds to psychological stressors by increasing corticosteroid release, Wirth, Welsh, and Schultheiss (2006) hypothesized that high-power individuals are more stressed by a defeat than are low-power individuals and that a defeat should manifest itself in cortisol release. McClelland (1989) likewise stressed the mediating influence that cortisol plays in organizing the general adaptation syndrome. Wirth et al. experimentally induced social defeat in a competitive contest. Participants worked on varying forms of a number tracking test. Half the participants experienced social defeat in this reaction-based cognitive task. The other half experienced social victory, that is, the task was so rigged that they “outperformed” the other participant. The experience of losing the contest against another surely qualifies as a power stress and, as predicted, its occurrence resulted in an increase in salivary cortisol secretion for persons scoring high in power motivation but not in those scoring low.

The process of evaluating a person for employment holds prospects for arousing power stress. As previously noted, power-motivated persons favor as friends and associates persons who are unassuming, unassertive, and disinclined to challenge the power-motivated person’s authority. Strong-willed persons, on the other hand, pose a distinct threat to the successful exercise of power. We obtained evidence to confirm this hypothesis (Fodor, Wick, & Hartsen, 2006). In a simulated employment interview, college males imagined themselves as manager to a young man playacting a videotaped interview. He portrayed himself as strong-willed and assertive in one condition, and compliant in another. The research participants who did the evaluating had scored either high or low on the PSE in *n* Power. As shown in Fig. 1.3, participants scoring high evinced higher electromyographic readings from the corrugator brow (frown) muscle when viewing the strong-willed, assertive candidate than did those scoring low, and higher than all participants who witnessed the compliant candidate. Responses to a scale that measured affect revealed the same pattern for negative affect.

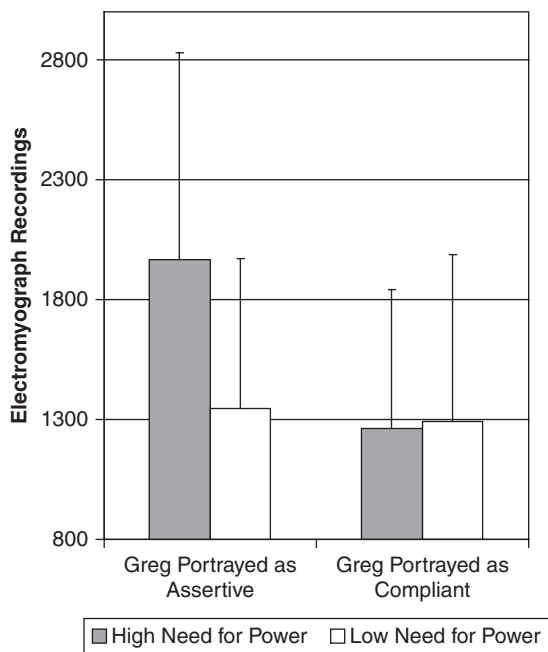


Figure 1.3 Mean EMG recordings from the brow supercilii muscles. Error bars represent standard errors of the mean. (Fodor, Wick, & Hartsen, 2006, p. 605.)

Creativity

McClelland (1964) put forth a theory about creative achievement in science that has a Freudian bent to it. Scientific achievement, he speculated, arises as a sublimation of repressed aggressiveness in childhood, assuming the form of power motivation:

It is certainly part of the folklore of science that it represents an attempt to *conquer* [italics original] nature, to dominate it and bring it under man's control. Social theorists have noted that the view that man could be more powerful than nature is by no means a common one in history. In fact, the usual belief among peoples of the world, even today, is that nature is more powerful than man . . . and that man must somehow placate the gods who control nature. It does not seem too farfetched to assume that it took an unusual psychodynamic situation to create in some men the apparently irrational belief that they could conquer nature. The blocked aggressive needs of a few scientists, diverted toward nature, may well have fulfilled such an important historic function. (pp. 171–172)

At a more personal level, we have noted that the scientist is intensely analytic in his approach to experience. His response is to freeze the flux of reality, to ask, What is

it?, to take the experience apart and see what makes it tick. It is no mere metaphor to say that the analysis represents a form of aggression. To take something apart is to destroy it in a very real sense. (p. 172)

Implicit in the power motive concept is the notion that a nonverbal incentive may consist of any behavioral opportunity that can result in a surge of positive affect associated with having had impact. Creativity, if its impact can be felt by others, easily qualifies as a nonverbal incentive for the power need. One thinks immediately of James Watson's autobiographical account of how in their quest to understand the molecular structure of DNA, he and Francis Crick were unabashedly driven by a desire to win the Nobel Prize and to be the first to go to press (Watson, 1968). Merton (1973) speaks of the "race for priority" as a major impetus to scientific discovery. Acclaim and recognition most adorn the scientist who is first to publish a scientific discovery. What these writings suggest is that an aspect of power motivation, namely, the desire to have impact (Eff in the Winter coding system) figures into the creative process as it pertains to scientific achievement.

We obtained experimental evidence that appears to support McClelland's (1964, 1975) view that power motivation may indeed result in creative performance (Fodor, 1990; Fodor & Carver, 2000; Fodor & Greenier, 1995). The research strategy required that college student participants at a technological university individually attempt to solve an engineering problem of moderate difficulty, namely, how to keep the classroom windows in an old building from getting so stuck during the winter months that they become difficult to open come spring. The experimenter then "examined" what the participant had written and provided written feedback couched in the language of positive or negative power imagery. Positive feedback informed the participant that the experimenter was impressed and that the participant's proposed solution likely would register a favorable impact on others, notably, members of the engineering and scientific community, essentially causing them to "stand up and take notice." Accompanying this handwritten feedback was a 7-point rating scale that conveyed the same message, that is, how much recognition and visibility the participant's solution was likely to command. The feedback thereby provided was credible because the window frame problem is sufficiently easy that anyone can render a solution that an evaluator might reasonably consider creative, yet the participant might just as likely think a more sophisticated solution possible. The negative feedback condition conveyed an opposite message, signifying that the proposed solution likely would not create much impact. A control condition entailed no experimental feedback.

Positive feedback stated in the language of power imagery enhanced creativity in high-power but not low-power persons, by two different measures: creativity ratings (Amabile, 1983) applied to proposed solutions to a second engineering problem (Fodor, 1990; Fodor & Carver, 2000) and scores on the

Remote Associates Test (RAT) of creativity (Mednick & Mednick, 1967; Fodor & Greenier, 1995), a test which Isen, Daubman, and Nowicki (1987) have shown to be a *state* measure that responds to conditions that instill positive affect. The format of the RAT consists of three words followed by a blank space, requiring that the participant provide in the blank space a word that relates to each of the three words. The following item is an example:

mower atomic foreign _____

The correct answer *power* is a remote, or uncommon, associate of the three stimulus words, far apart from them in semantic space. The RAT has relevance to the study of creativity in science and engineering. Gordon (reported in Mednick & Mednick, 1967) studied scientists and engineers engaged in defense-related research and found that those who scored high on the RAT wrote a greater number of successful research proposals than did those who scored low. Table 1.2 displays the results we obtained with the RAT in one of the studies we did (Fodor & Greenier, 1995).

Negative feedback, phrased in the language of power imagery, produced distinctly low creative outcomes by both of the aforementioned measures in high-power individuals. Measures of positive versus negative affect subsequent to feedback displayed the same pattern, suggesting that creativity and affect levels go hand in hand. The patterns we observed for high-power individuals did not occur in persons scoring low in power motivation, who were relatively unaffected by the feedback they received. What the power-motivated individual perceives as creative success, therefore, appears to manifest itself in both good feeling and creative performance. The experience of failure to register impact through their creative endeavors appears to constitute a power stress in power-motivated persons and to diminish creative performance at least temporarily immediately thereafter. The “sweet smell of success,” if it entails recognition and admiration, may well be a guiding motif in the creative life of the power-motivated person, giving sustenance to the power need.

Table 1.2 Remote Associates Test Scores (Fodor & Greenier, 1995, p. 248)

	Positive Feedback	Negative Feedback	No Feedback
High <i>n</i> Power			
<i>M</i>	16.4 (n = 22)	10.7 (n = 22)	12.1 (n = 22)
<i>SD</i>	3.8	4.5	3.8
Low <i>n</i> Power			
<i>M</i>	12.3 (n = 22)	11.1 (n = 22)	11.6 (n = 22)
<i>SD</i>	3.8	4.6	3.6

Note: From “The power motive, self-affect, and creativity,” by E. M. Fodor and K. D. Greenier, 1995, *Journal of Research in Personality*, 29, p. 248.

Other Validity Correlates of *n* Power

There is an abundant research literature to support the construct validity of the PSE measure for *n* Power (McAdams, 2006; Winter, 1996). Much of the discussion in this chapter speaks to the validity of the *n* Power concept and the PSE measure by which it is assessed. As previously mentioned, persons high in the power motive are especially responsive to ingratiation by a subordinate (Fodor & Farrow, 1979). This finding strengthens the construct validity of the PSE measure of *n* Power insofar as it is consistent with Karen Horney's (1937) view that persons needful of power and prestige have experienced a lack of parental affection in their early years. Ingratiation from a subordinate logically compensates for that lack in their lives. Also, members of small groups perceive high-*n*-Power leaders as dominant. Moreover, the presence of a leader high in the power motive appears to inhibit information flow into group discussion (Fodor & Smith, 1982) and reduces feelings of competence in group members (Fodor & Riordin, 1995), presumably because power-motivated persons dominate others and inhibit their self-expression. These findings tie together with the theoretical assumptions that underlie the *n* Power concept.

Mason and Blankenship (1987) studied the relationship between *n* Power and the infliction of physical abuse in intimate relationships. They employed a modified version of the Conflict Tactics scales (Straus, 1979) to measure the amount of psychological and physical abuse college students had inflicted on their partners in the past year. High-*n*-Power men (but not women) inflicted significantly greater abuse on their partners than did low-*n*-Power men. Zurbriggen (2000), correspondingly, obtained positive correlations between *n* Power in men (but not women) and the Coerce and Seduce scales that she administered to test aspects of aggressive sexual behavior.

Further evidence for the validity of implicit measures for *n* Power comes from the finding that *n* Power correlates with frequency of sexual intercourse (McClelland, 1975; Schultheiss, Dargel, & Rohde, 2003; Winter, 1973). Schultheiss et al. (2003) suggest that these findings neatly parallel evidence that social dominance coincides closely with sexual reproduction in other mammalian species. Citing Wilson (1980), they point out that dominance rank among social mammals confers privileged access to mates with a resultant increase in offspring by comparison with mammals of subordinate rank. In humans, the reasoning goes, the implicit power motive (as measured by the PSE) may serve as a marker for actual or desired dominance rank and may be related to reproductive success. Power-motivated individuals show high baseline levels of testosterone, a hormone associated with sexual

motivation in both men and women, and they also evince a greater increase in testosterone level when winning a dominance contest than do low-*n*-Power individuals (Schultheiss, Campbell, & McClelland, 1999; Schultheiss & Rohde, 2002).

Looking to the Future

One can sense an interplay between motivation as measured by the PSE and people's reactions to emotional cues provided by other persons. These cues can serve as nonverbal incentives in relation to *n* Affiliation and *n* Power. Consistent with McClelland's (1989) theoretical analysis of the role nonverbal incentives play in activating implicit motives, Schultheiss and Hale (2007) examined how power- and affiliation-motivated individuals oriented toward faces denoting various emotions, specifically, their allocation of attention to the motive-relevant cues inherent in facial expression. They presented participants with faces known to express the emotions of joy, anger, and surprise relative to a neutral expression. Prior research has consistently shown that people can rate facial expressions on the degree to which they reflect dominance (power) and affiliation (e.g., Knutson, 1996). Joyful expressions rate as high on both dominance and affiliation, and anger expressions rate high on dominance but low on affiliation. As regards power motivation, Schultheiss and Hale reasoned that the incentive value of an emotional expression connoting dominance should be *reciprocally* related to the dominance the facial expression displays because such an expression from another signals lack of control by the power-motivated person. Schultheiss and Hale hypothesized, therefore, that facial expressions of high dominance, notably, anger and joy, should figure as *disincentives* in the minds of power-motivated persons, cues they would orient their attention away from. Surprise, as a low-dominance expression, should by contrast serve as an *incentive* for the power-motivated person, giving evidence that the person has produced an impact.

The method by which Schultheiss and Hale tested their predictions was Mogg and Bradley's (1999) dot-probe task. The computer screen presents an emotional and a neutral face side by side. The stimuli then are masked, and a little dot (the probe stimulus) replaces either the emotional or the neutral face just previously seen. How fast the participant orients toward the probe signifies how much the individual's attention was oriented toward the facial stimulus the dot replaces, be it emotional or neutral. The response latency is shorter if the participant's attention was focused on the face that the dot replaces, longer if attention was toward the face that was not replaced by the dot. In line with prediction, the authors found that power-motivated

individuals attended to the low-dominance, and therefore, rewarding surprise expression but oriented away from the high-dominance, and therefore aversive, anger and joy facial expressions. These findings were in sharp contrast to what the authors saw in affiliation-motivated participants. The most prominent finding for affiliation-motivated persons was that they showed high vigilance for anger faces, suggesting a sensitivity to interpretational cues signaling rejection.

Exploring the question of emotional reactivity to other persons, we have begun a simulated dating service experiment. College student participants scoring high or low in *n* Power observe an 8-minute playacted video depicting an interview with an opposite-sexed person who comes across as either assertive or deferentially compliant. Outcome measures consist of electromyographic readings from the brow corrugator (frown) muscle and responses to two different questionnaires that assess affective response. The prediction, of course, is that high-*n*-Power males, consistent with Winter's early findings, should show an aversive reaction to assertive women, preferring women who are more deferent to the male species. As regards women vis-à-vis men, we anticipate a similar finding, although frankly not with great confidence. There is reported evidence that women factor male dominance into their assessment of a male's desirability to so significant a degree that physical appearance plays less a role in their expressed preference than it does with males evaluating women (cf. Cunningham, Barbee, & Pike, 1990). The question is: Does *n* Power moderate women's preference for dominant males?

We can conclude from the foregoing discussion that the *n*-Power concept is embedded within an intricate and expanding nomological network. What appears to have given sustenance to the concept and kept it alive through the years is its inherently multifaceted nature, which has enabled research scholars to pursue its meaning into realms as diverse as physiology, cognition, group dynamics, creativity, and considerations of mental health. As other writings in the present volume amply demonstrate, *n* Power also explains important aspects of leadership performance. In summary, the *n* Power concept has enhanced our understanding of many dimensions of human behavior that without the research the concept has spawned would remain largely unexplained.

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Chapter 2

The Achievement Motive: A Review of Theory and Assessment of *n* Achievement, Hope of Success, and Fear of Failure

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Implicit achievement motivation, often denoted *n* Achievement, refers to a nonconscious and recurrent preference for affectively rewarding experiences related to improving one's performance (Atkinson, 1957). In the following chapter, I will provide a brief and targeted history of the development of the construct of implicit achievement motivation, focusing on the legacy of David McClelland and John Atkinson. The first section highlights the major theoretical and methodological landmarks in achievement motivation research under the McClelland-Atkinson tradition and will cover such topics as measurement, developmental considerations, motivational incentives, and behavioral outcomes. I will also survey some of the major content areas in which implicit achievement motivation researchers have undertaken work in the past. Finally, one of the areas that has not received sufficient attention is the distinction between hope of success (HS) and fear of failure (FF) motivation, particularly in the development of valid measures of these motives, and I end the chapter with a description of and validation data for a new *n* Achievement measure that takes this HS/FF distinction into consideration.

McClelland's Legacy of Achievement Motivation Research

Briefly, David McClelland's legacy is divided into three main areas, the benefits of which extend not only toward *n* Achievement research but also to implicit motivation research in general:

One, his research program influenced a group of researchers with members such as Atkinson, Feather, Koestner, and Winter, who adopted his ideas and research principles so as to develop them further into the full program of work that exists today. For instance, McClelland's (1961) argument that the preferences of *n* Achievement motivated people for activities that involve skill and effort, provide moderate challenge and risk, and present clear performance feedback are more likely to be satisfied in entrepreneurial situations, spawned a formidable body of research on the relationship between *n* Achievement and entrepreneurial behavior and economic productivity (c.f., Collins, Hanges, & Locke, 2004). In addition, Atkinson, a researcher who was directly influenced by McClelland, developed an expectancy-value theory of *n* Achievement that has since led to many productive findings and subsequent theory making about the nature of *n* Achievement (see Cooper, 1983; McClelland, 1987; Schultheiss & Brunstein, 2005).

Two, starting with the publication of *The Achievement Motive* (McClelland, Atkinson, Clark, and Lowell, 1953), McClelland and his colleagues popularized a technique of measuring motivation scientifically, through first experimentally arousing the relevant motive and then developing content coding systems that were based on the effects of motive arousal. McClelland's introduction of these assessment procedures gave researchers ways of measuring *n* Achievement (and other motives) accurately, providing a powerful methodology with which to conduct and evaluate subsequent research. The publication of *The Achievement Motive* also led to a theory about the structure of the achievement-motivated process, which will be described later.

The third accomplishment from McClelland's long history of work is his championship of the distinction between implicit and explicit/self-attributed motivation. McClelland, Koestner, and Weinberger (1989) argued convincingly that these two types of motivation are distinct, each with their own incentives, behavioral outlets, and correlates. The eventual acceptance of the implicit-explicit motive distinction allowed motivation researchers to explain findings that were previously difficult to explain, specifically clarifying many otherwise conflicting characteristics of *n* Achievement.

Thus, I will frame the first part of this literature review around McClelland's three research contributions. I will start with his theoretical concept of *n* Achievement as it is described in *The Achievement Motive* as well as the research and assessment tradition that resulted from this work, chiefly, the development of the theory of *n* Achievement after Atkinson's (1957) expectancy-value theory of *n* Achievement. Finally, I explain why the implicit-explicit distinction calls for a revision of *n* Achievement measures and theory.